

MULTIOBJECTIVE OPTIMIZATION FOR TOPOLOGY OF HETEROGENEOUS WIRELESS SENSOR NETWORKS

Consolidated Results

Table 1: ϵ values for each SAP instance.

Instance size	Density (D)	ϵ values
10 x 10	10%	9, 18, 27, 36, 45, 54, 63, 72, 81, 90
	20%	10, 20, 30, 40, 50, 60, 70, 80, 90, 100
	30%	10, 20, 30, 40, 50, 60, 70, 80, 90, 100
	40%	9, 18, 27, 36, 45, 54, 63, 72, 81, 90
	50%	10, 20, 30, 40, 50, 60, 70, 80, 90, 100
15 x 15	10%	11, 22, 33, 44, 55, 66, 77, 88, 99, 110, 121, 132, 143, 154, 165
	20%	14, 28, 42, 56, 70, 84, 98, 112, 126, 140, 154, 168, 182, 196, 210
	30%	13, 26, 39, 52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195
	40%	14, 28, 42, 56, 70, 84, 98, 112, 126, 140, 154, 168, 182, 196, 210
	50%	15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180, 195, 210, 225
20 x 20	10%	13, 26, 39, 52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195, 208, 221, 234, 247, 260
	20%	20, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 320, 340, 360, 380, 400
	30%	18, 36, 54, 72, 90, 108, 126, 144, 162, 180, 198, 216, 234, 252, 270, 288, 306, 324, 342, 360
	40%	20, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 320, 340, 360, 380, 400
	50%	19, 38, 57, 76, 95, 114, 133, 152, 171, 190, 209, 228, 247, 266, 285, 304, 323, 342, 361, 380

Table 2: Configuration parameters for validation scenarios of the proposed model.

Parameter	Description	Values
$w = h$	Size of the square monitoring region (in points)	{10 , 15 , 20}
D	Density of transceiver allocation points (in %)	{10 , 20 , 30 , 40 , 50}
Z	Scale of the monitoring region (in meters)	800
$time$	Maximum execution time for each ϵ value (in hours)	12
gap	Optimization gap tolerance (in %)	0.1

Table 3: Numerical results for instance of size 10×10 and $D = 50\%$.

ϵ	Allocated nodes (S2C)	Allocated nodes (S2CPro)	Objective I	Objective C	Gap (%)	Time in Solver (s)	Total Time (s)
10	3	0	0.9176	12	0	1.8163	6.31
20	4	0	1.2235	21	0	5.4797	9.72
30	2	1	1.3676	41	0	7.2536	11.63
40	2	1	1.3676	40	0	1.8715	6.06
50	2	1	1.3676	53	0	1.7859	5.83
60	3	1	1.6735	61	0	23.549	27.64
70	1	2	1.8176	70	0	21.4564	25.43
80	2	2	2.1235	80	0	27.0705	31.02
90	1	3	2.5735	90	0	20.1489	24.28
100	1	4	3.3894	100	0	13.735	17.87

Number of positions eligible for allocation: 50.

Binary variables: 30600. Integer variables: 30652.

Total of variables: 30652. Total of constraints: 61055.

Table 4: Numerical results for instance of size 15×15 and $D = 50\%$.

ϵ	Allocated nodes (S2C)	Allocated nodes (S2CPro)	Objective I	Objective C	Gap (%)	Time in Solver (s)	Total Time (s)
15	6	0	1.8353	21	0	15.29	40.15
30	6	0	1.8353	30	0	12.57	37.16
45	5	1	2.2853	54	0	562.7	586.8
60	5	1	2.2853	64	0	20.64	44.54
75	2	1	2.5912	76	11.49	43200.3	43223.33
90	4	2	2.7353	90	0	19210.9	19246.41
105	4	2	2.7353	105	0	82.74	106.02
120	3	3	3.1853	138	0	25367.38	25391.21
135	3	3	3.1853	135	0	597	620.88
150	2	4	3.6353	151	0	25574.81	25597.88
165	3	4	3.9412	167	0.0816	20882.31	20904.42
180	2	5	4.3912	182	0	24025.52	24049.83
195	1	6	4.8412	195	0	10171.03	10194.87
210	3	6	5.453	210	0	3967.6	3992.83
225	-	-	-	-	-	43200	43223

Number of positions eligible for allocation: 112.

Binary variables: 152099. Integer variables: 152213.

Total of variables: 152213. Total of constraints: 303642.

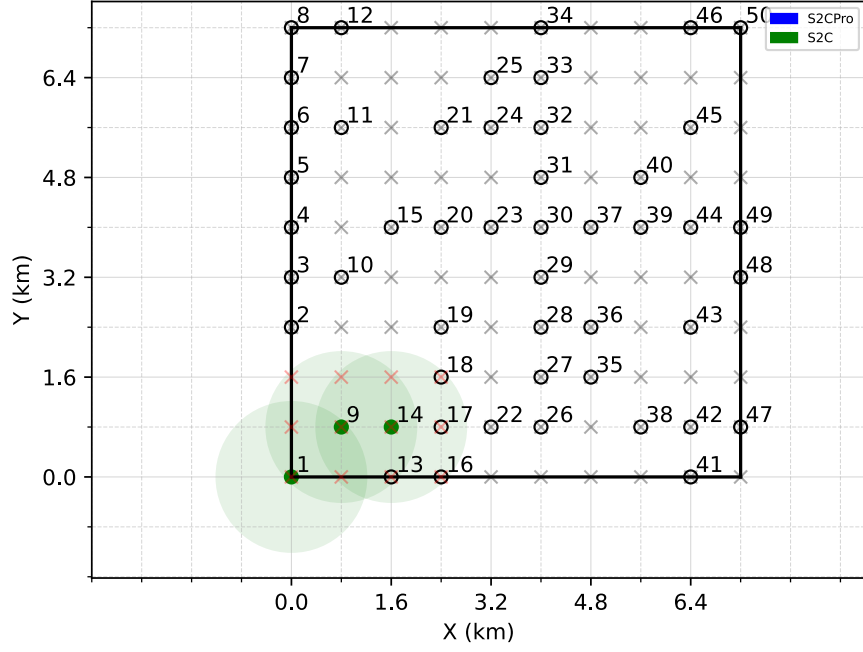
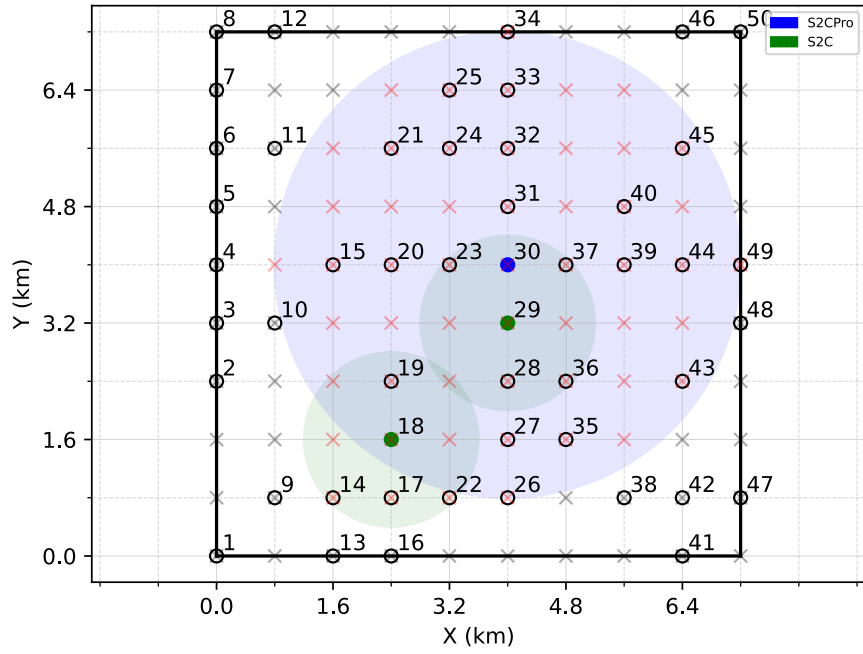
Table 5: Numerical results for instance of size 20×20 and $D = 30\%$.

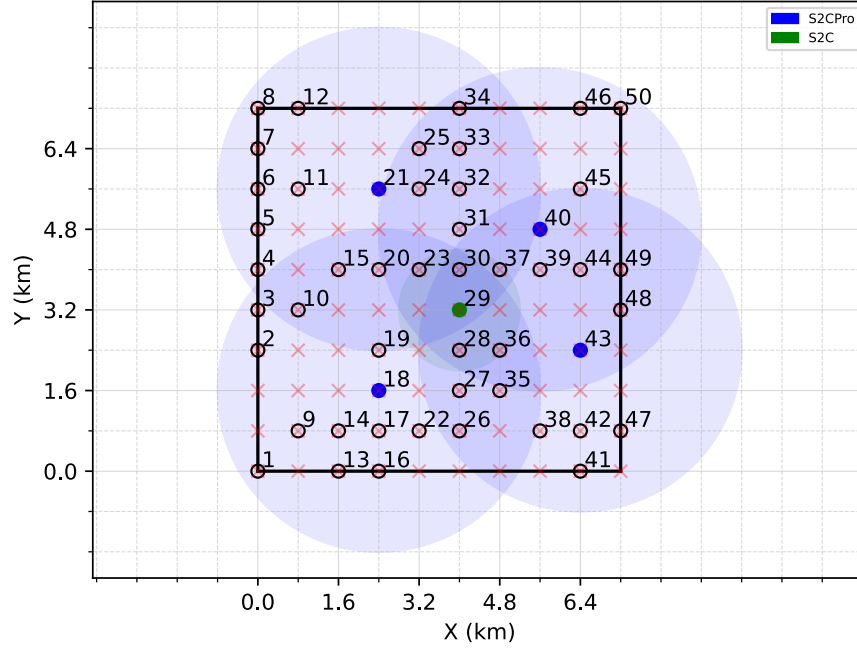
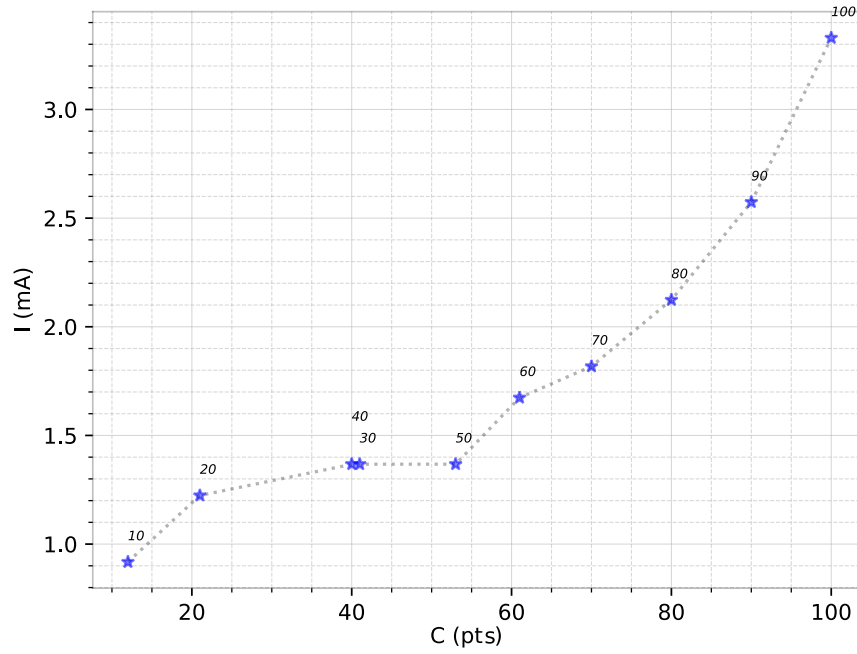
ϵ	Allocated nodes (S2C)	Allocated nodes (S2CPro)	Objective I	Objective C	Gap (%)	Time in Solver (s)	Total Time (s)
18	6	0	1.8353	20	0	15.49	67.43
36	7	0	2.1412	37	0	258.88	288.05
54	5	1	2.2853	57	0	35.11	64.02
72	5	1	2.2853	72	0	383.86	413.35
90	4	2	2.7353	90	0	17290.51	17321.14
108	4	2	2.7353	108	0	213.08	243.36
126	3	3	3.1853	127	5.92	43200	43229
144	3	3	3.1853	144	0	246.41	283.36
162	2	4	3.6353	162	0	34488	34518
180	1	5	4.0853	183	8.5	43200	43229
198	2	5	4.3912	198	5.23	43200	43235
216	2	6	5.1471	216	14.68	43200	43246
234	2	7	5.903	234	17.63	43200	43238
252	3	7	6.2089	255	14.56	43200	43237
270	3	8	6.9648	273	15.76	43200	43233
288	2	9	7.4148	289	13.59	43200	43242
306	4	10	8.7824	306	19.99	43200	43233
324	1	12	9.3765	325	16.87	43200	43233
342	2	13	10.4383	342	14.86	43200	43230
360	5	14	360	12.1119	13.11	43200	43230

Number of positions eligible for allocation: 120.

Binary variables: 213120. Integer variables: 213242.

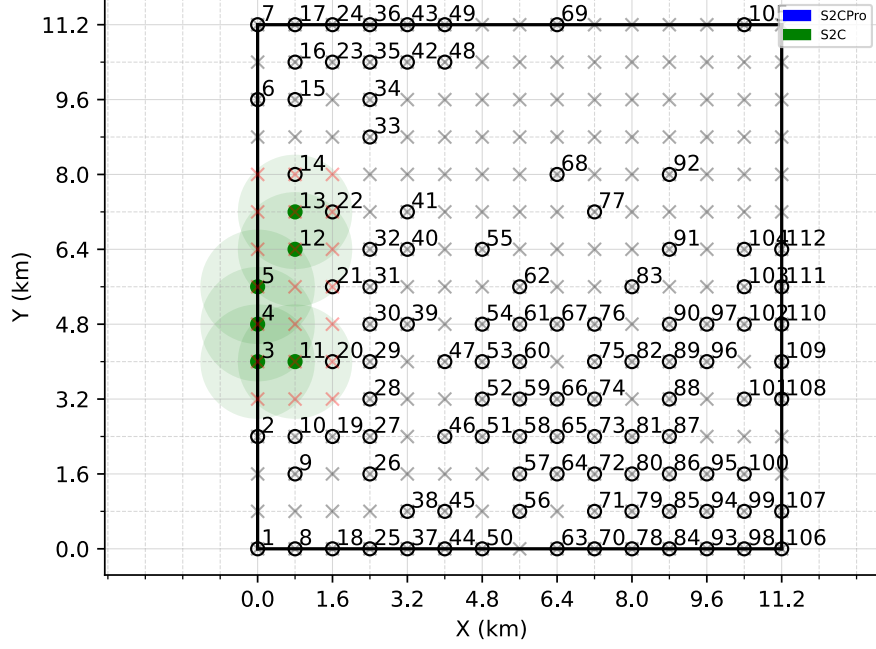
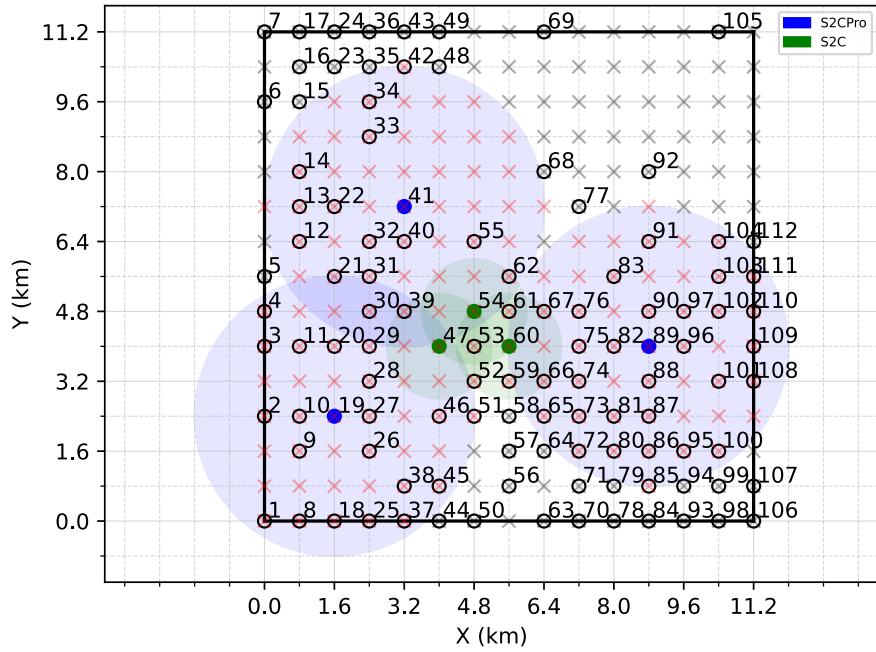
Total of variables: 213242. Total of constraints: 387325.

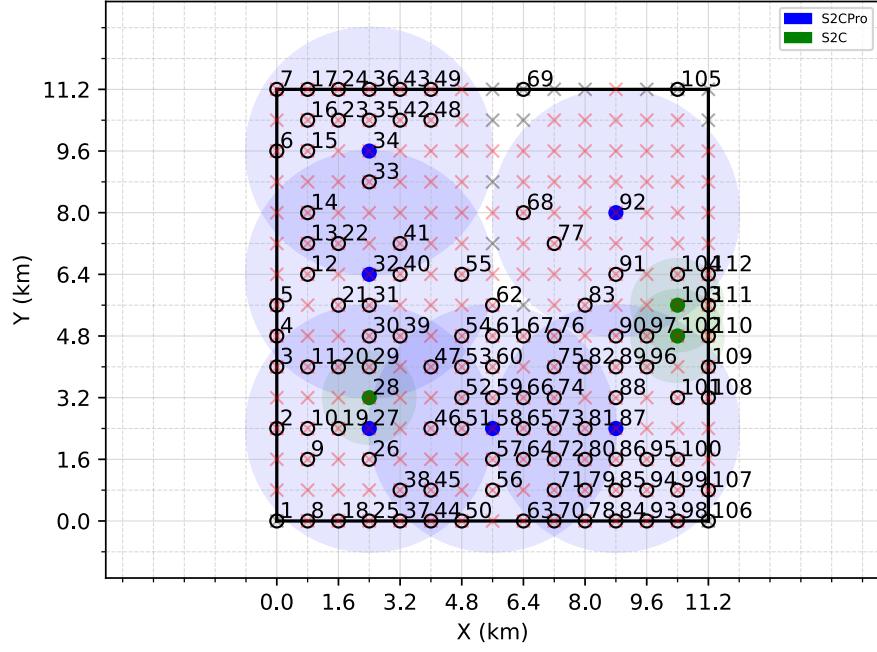
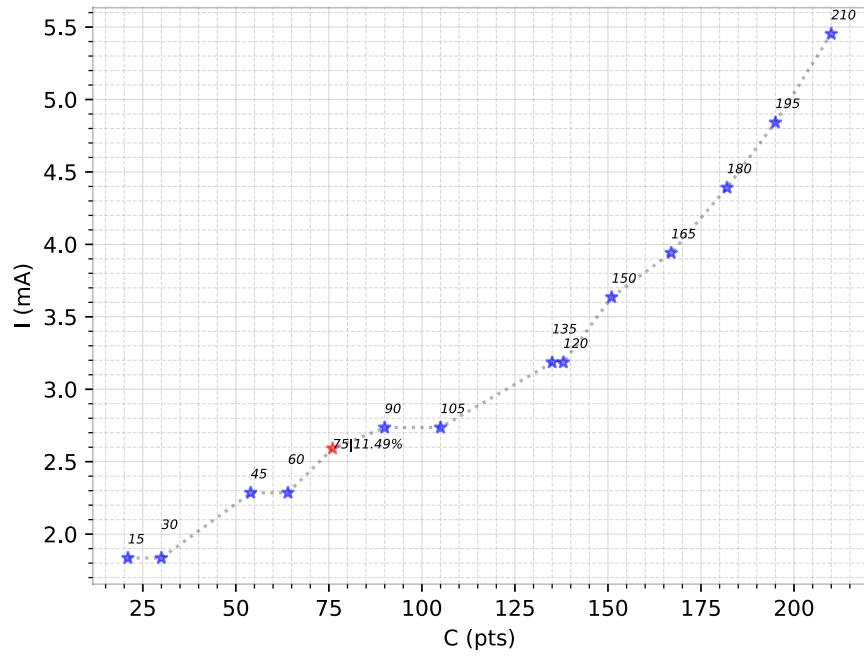
(a) $\epsilon = 10$ (b) $\epsilon = 50$ **Figure 1:** Best results for instance of size 10×10 and $D = 50\%$, for various values of ϵ .

(c) $\epsilon = 100$ 

(d) Pareto Frontier (approximate)

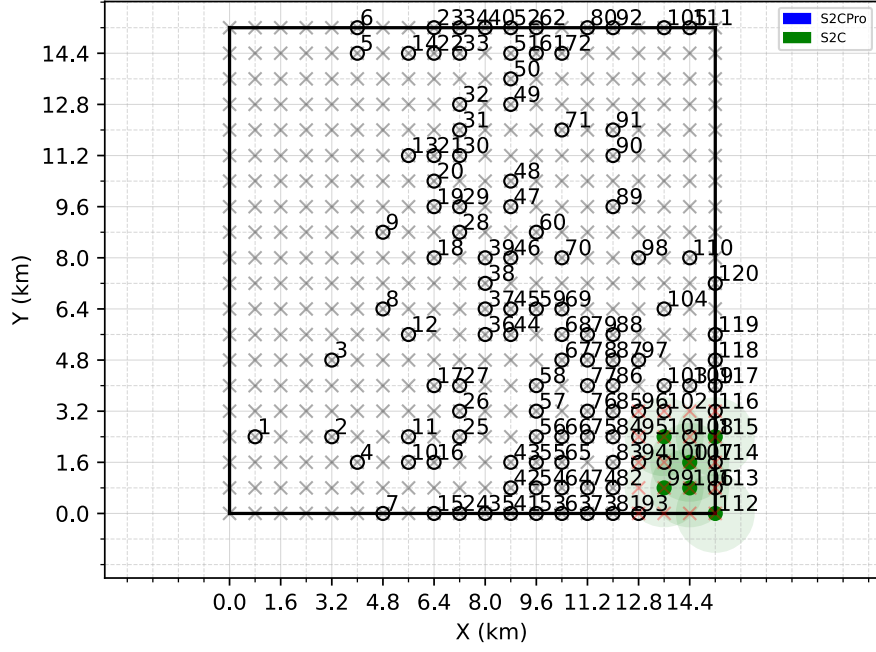
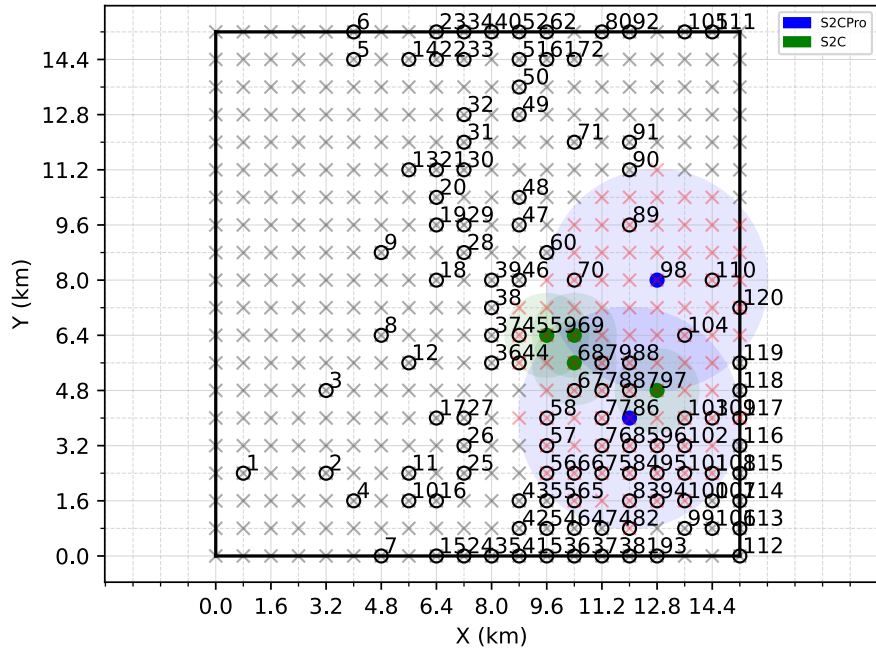
Figure 1: Best results for instance of size 10×10 and $D = 50\%$, for various values of ϵ .

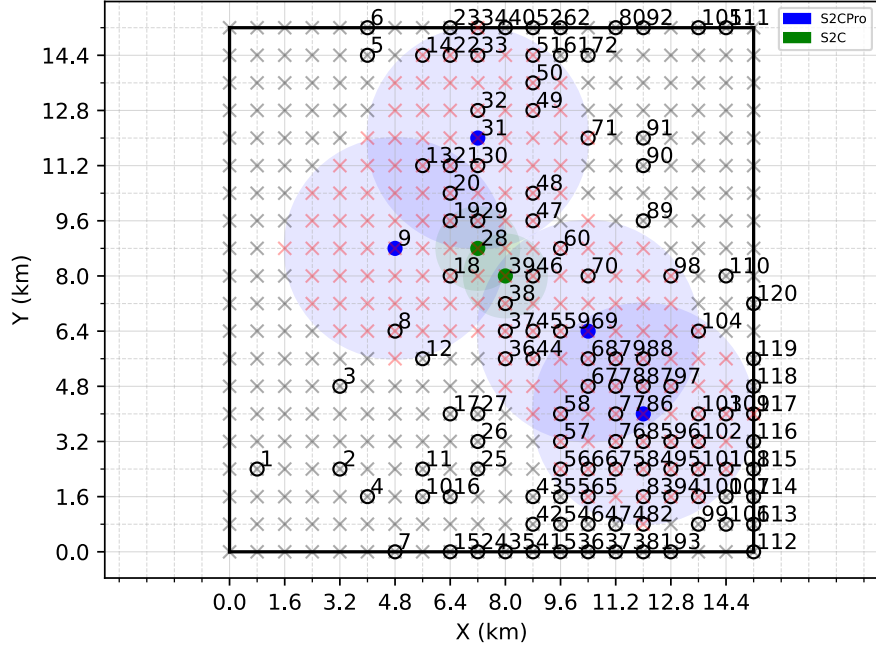
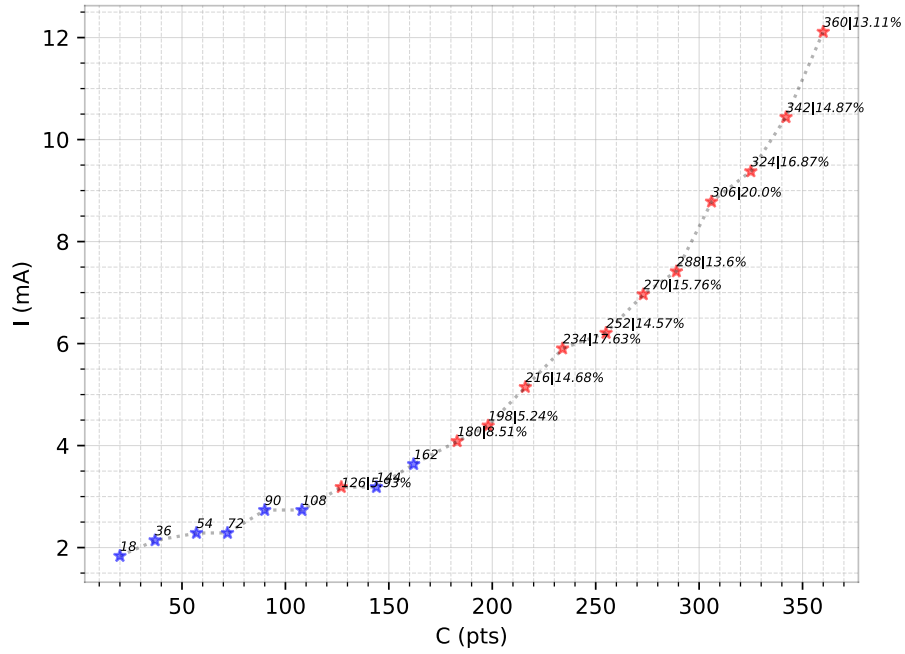
(a) $\epsilon = 15$ (b) $\epsilon = 120$ **Figure 2:** Best results for instance of size 15×15 and $D = 50\%$, for various values of ϵ .

(c) $\epsilon = 210$ 

(d) Pareto Frontier (approximate)

Figure 2: Best results for instance of size 15×15 and $D = 50\%$, for various values of ϵ .

(a) $\epsilon = 18$ (b) $\epsilon = 90$ **Figure 3:** Best results for instance of size 20×20 and $D = 30\%$, for various values of ϵ .

(c) $\epsilon = 162$ 

(d) Pareto Frontier (approximate)

Figure 3: Best results for instance of size 20×20 and $D = 30\%$, for various values of ϵ .

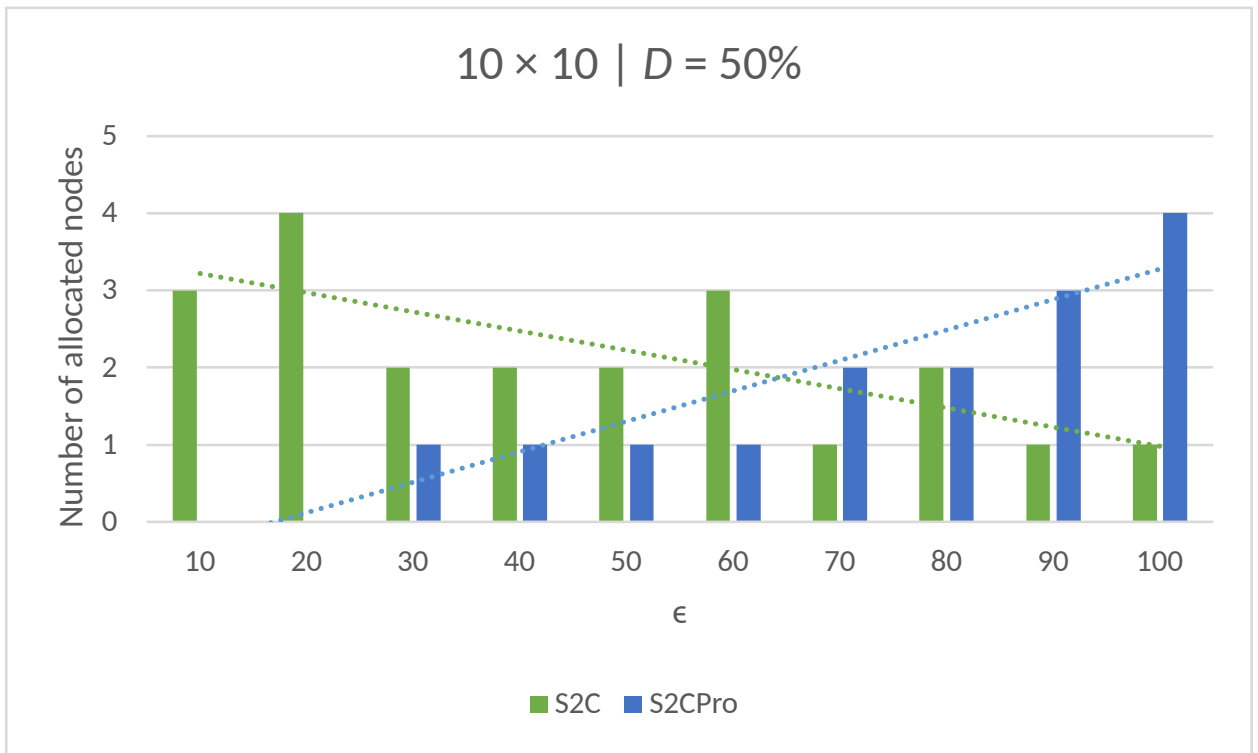


Figure 4: Number of allocated transceiver types by variation of ϵ for instance of size 10×10 and $D = 50\%$.

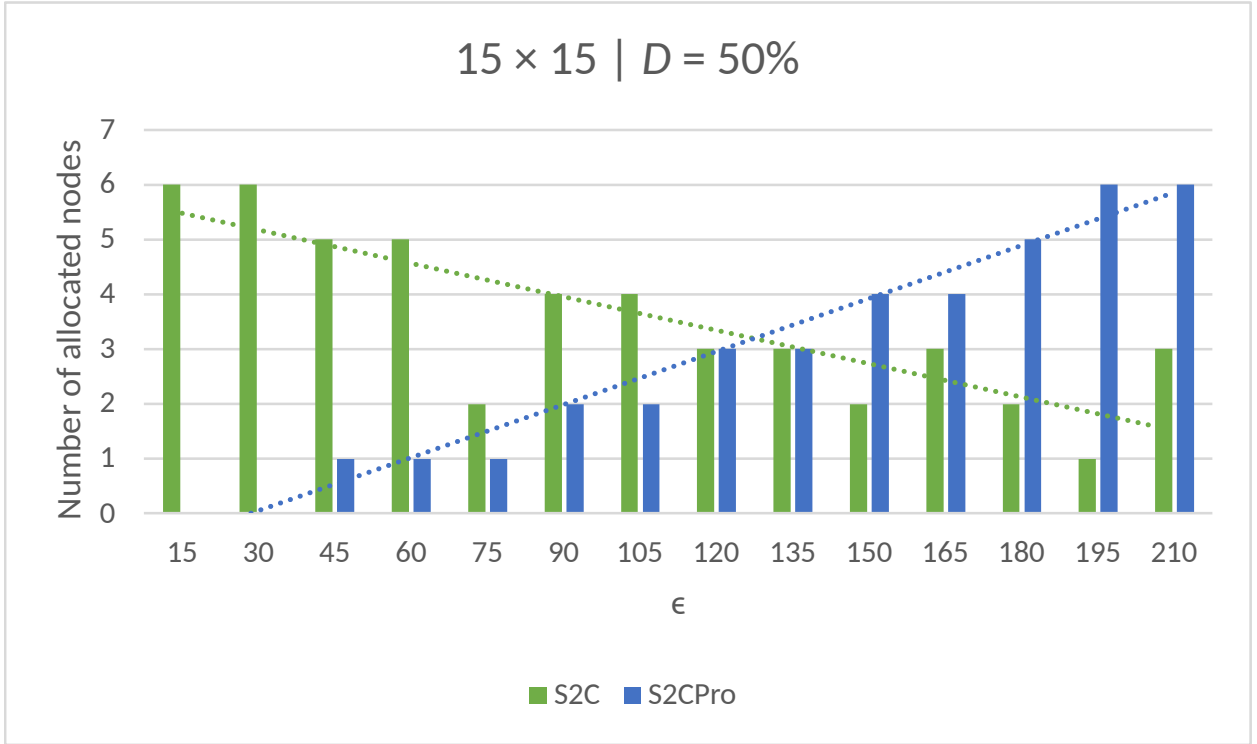


Figure 5: Number of allocated transceiver types by variation of ϵ for instance of size 15×15 and $D = 50\%$.

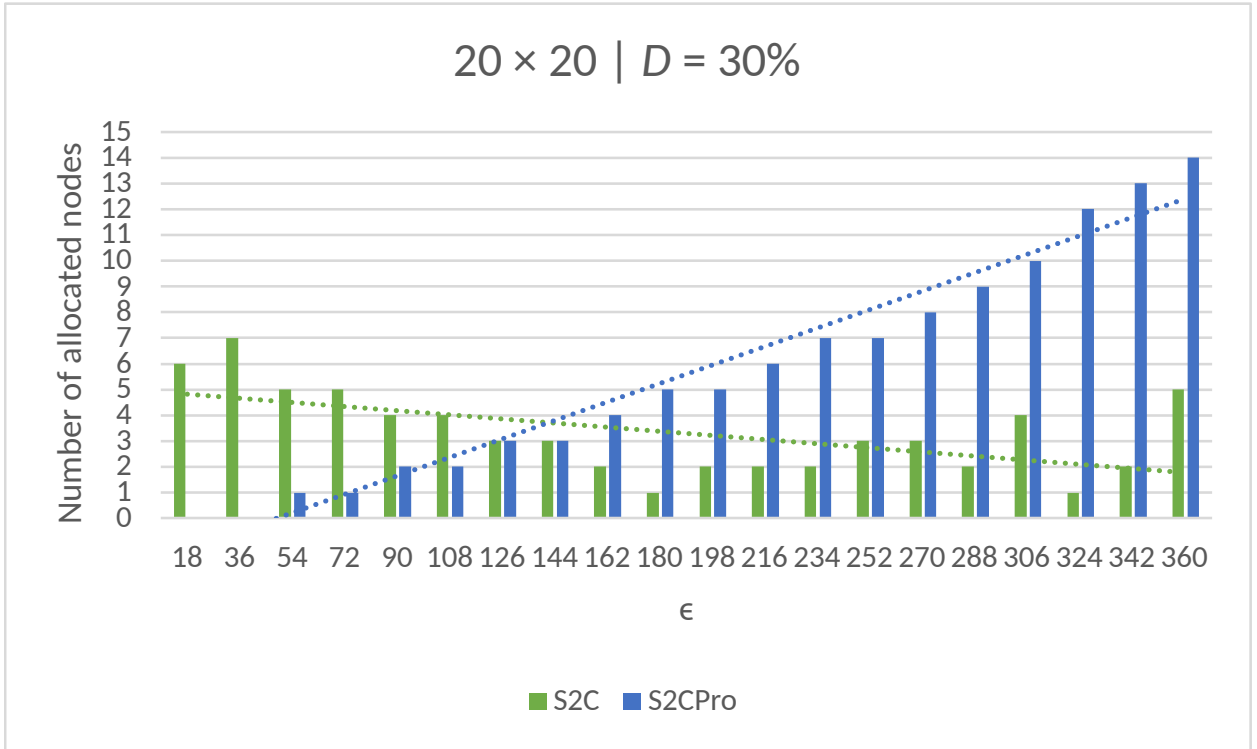


Figure 6: Number of allocated transceiver types by variation of ϵ for instance of size 20×20 and $D = 30\%$.

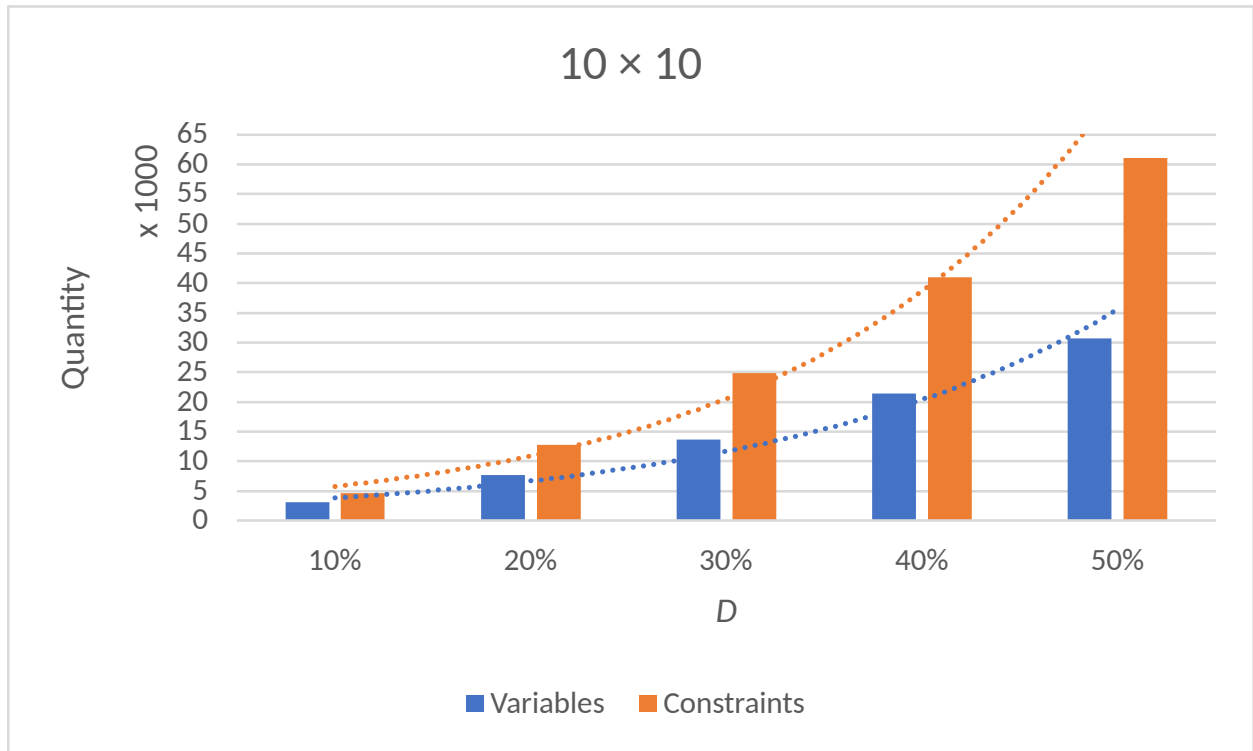


Figure 7: Number of variables and constraints for instances of size 10×10 .

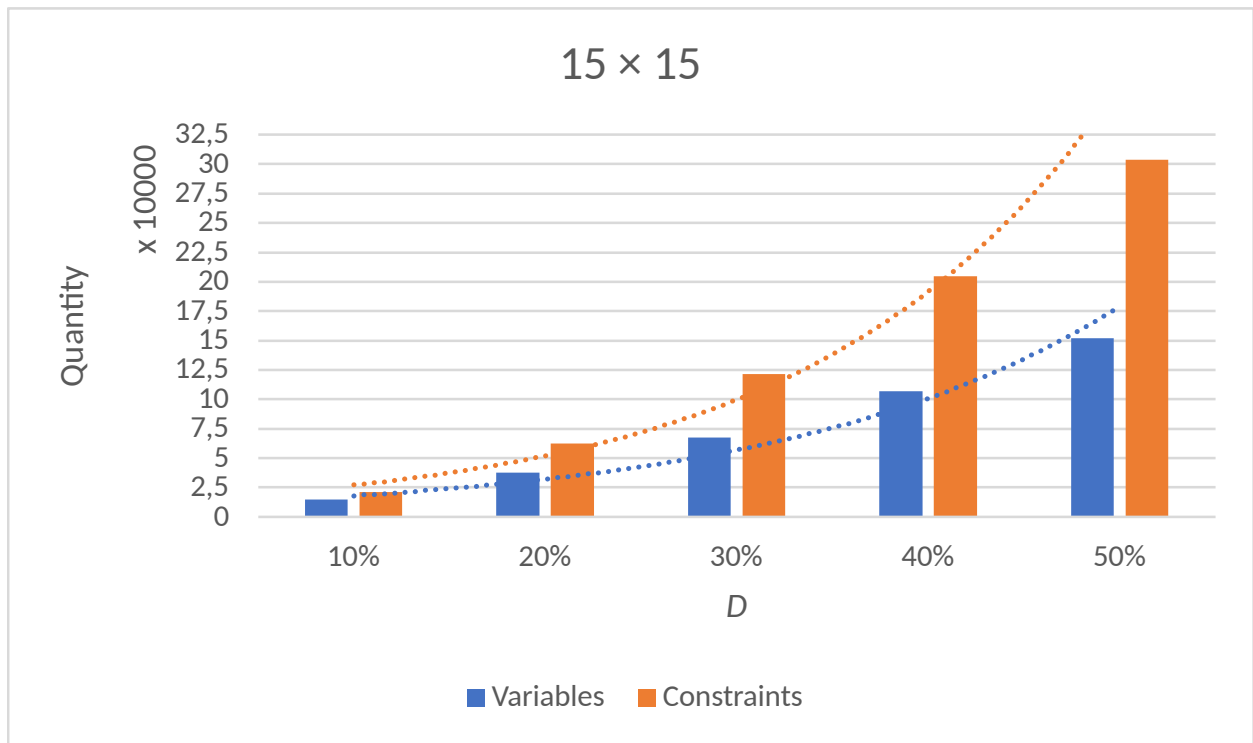


Figure 8: Number of variables and constraints for instances of size 15×15 .

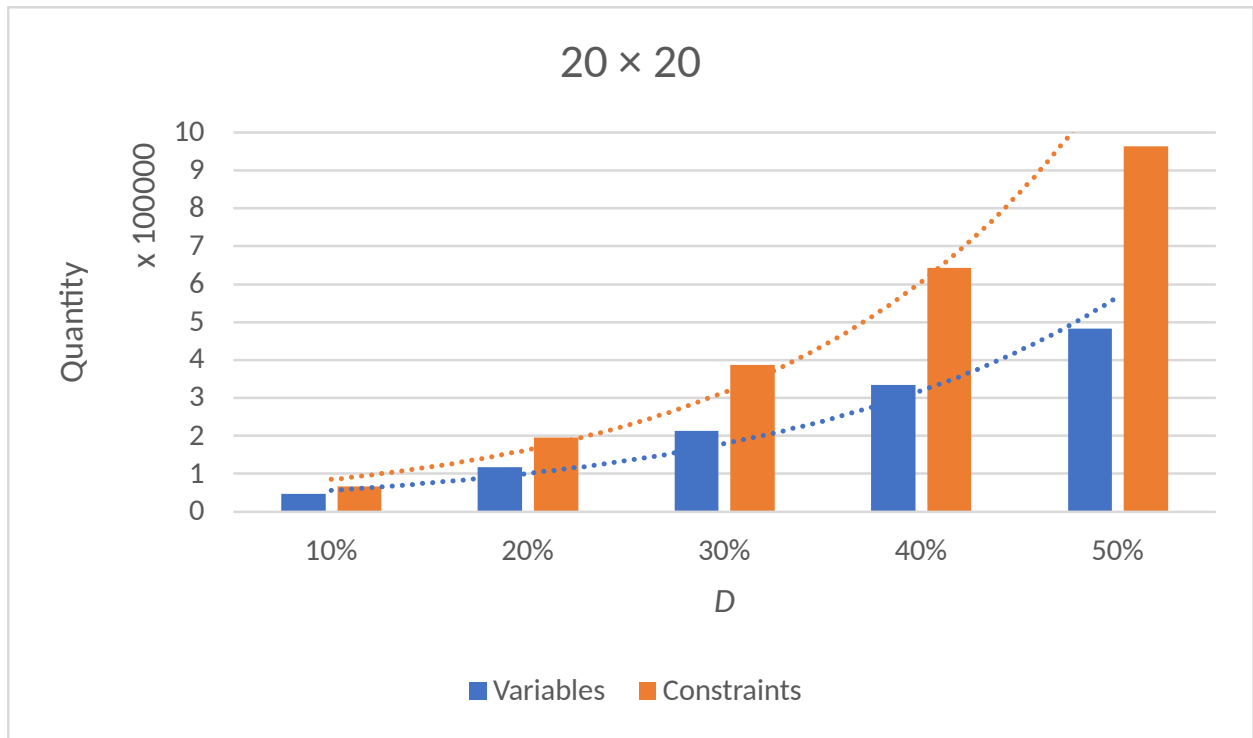


Figure 9: Number of variables and constraints for instances of size 20×20 .

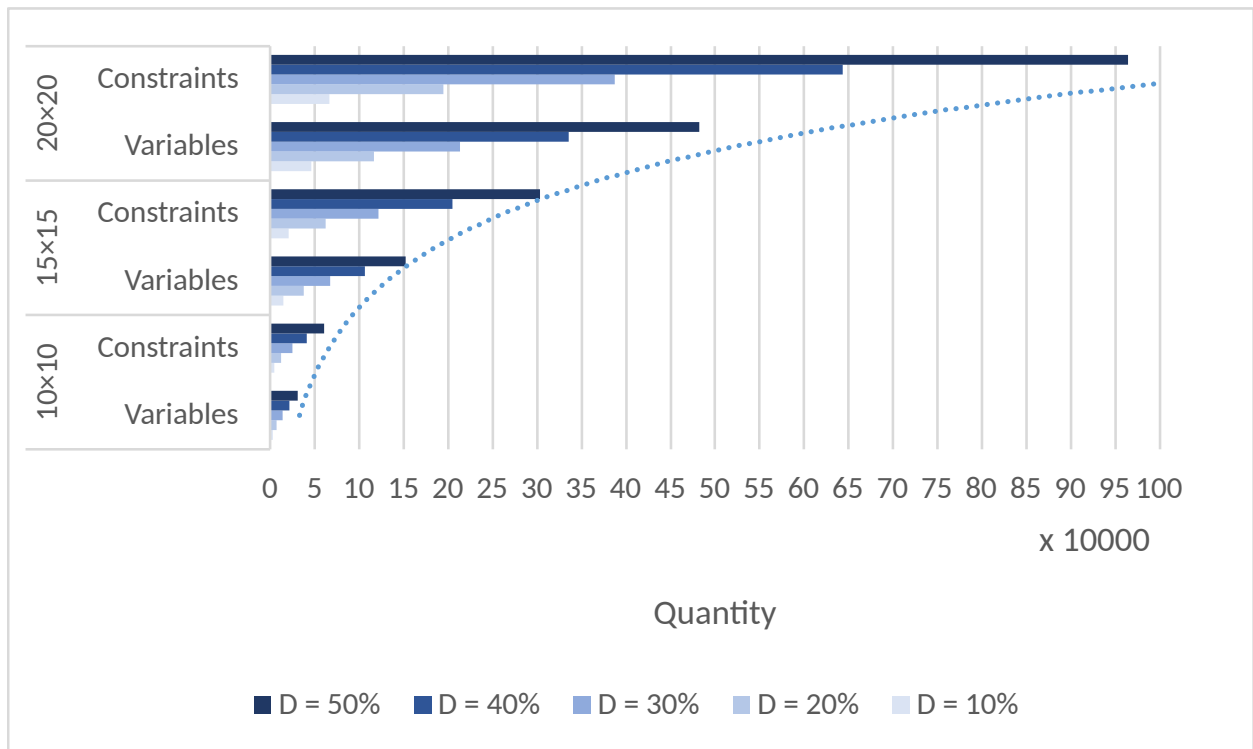


Figure 10: Number of variables and constraints for all instances.

Figure 11: Execution times for each ϵ value of instance size 10×10 and $D = 50\%$.

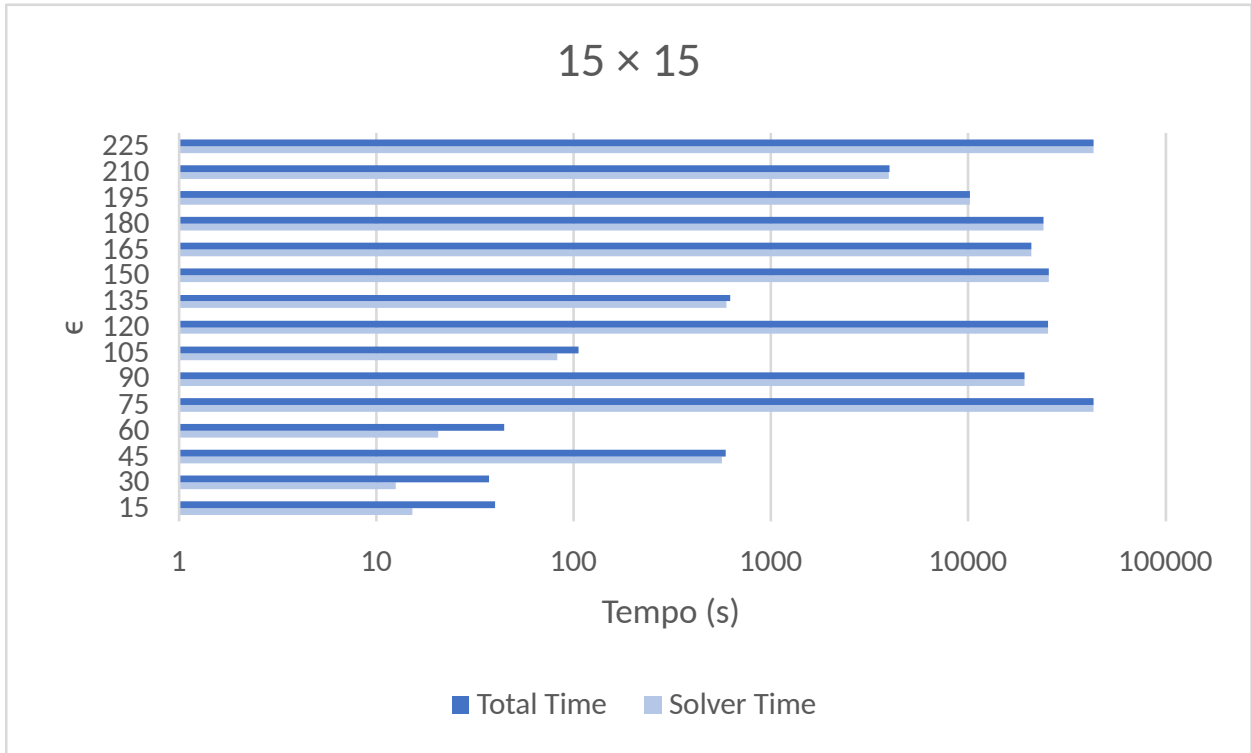


Figure 12: Execution times for each ϵ value of instance size 15×15 and $D = 50\%$.

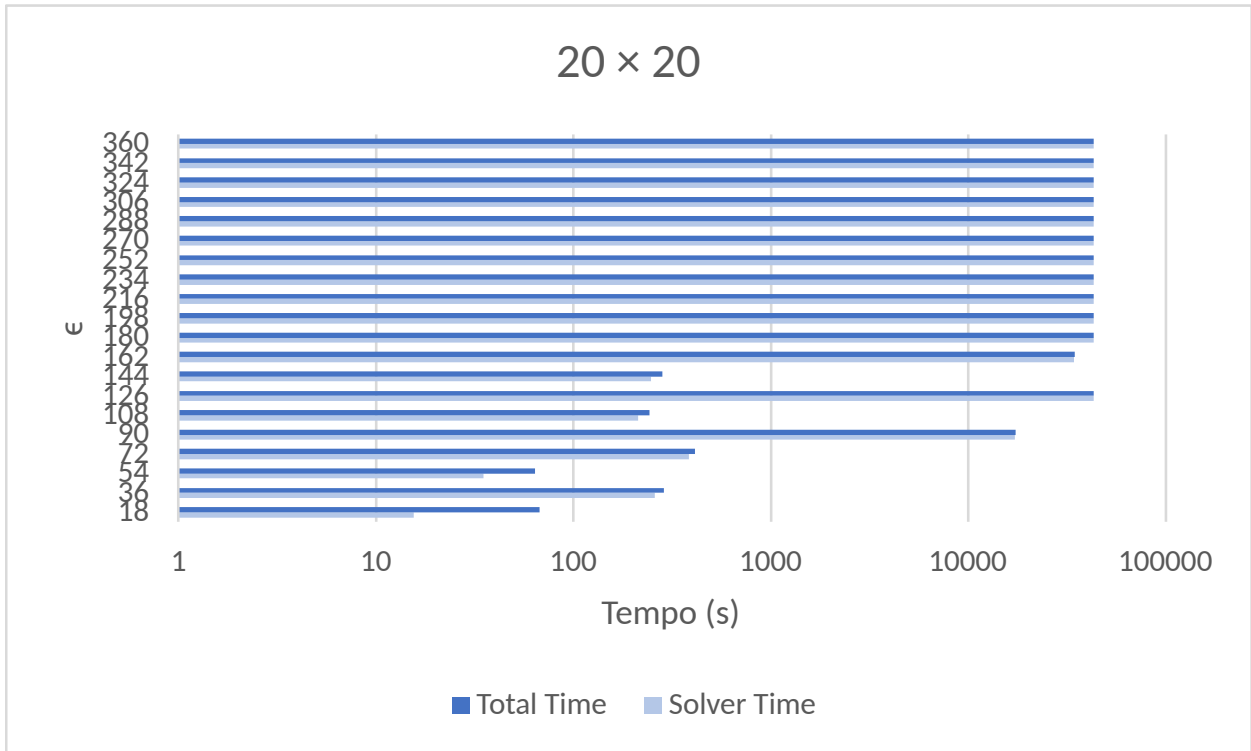


Figure 13: Execution time for each ϵ value of instance size 20×20 and $D = 30\%$.